

Ex04A

- Due No Due Date
- Points 100
- Submitting an external tool
- Attempts 0
- Allowed Attempts 1

← Attempt 1

Date Started

September 22, 2026 2:52 pm

Date Completed

September 22, 2026 2:52 pm

Time Spent

0m 5s

Score

Waiting on Instructor

☐ Hide correct responses

Please set your browser's zoom level to 100% for any exercise, homework, or exam that uses Boolean operations/equations.

Several browsers (such as Chrome) tend to use 90% zoom by default. Some browsers at non-100% zoom levels do not correctly render the "overbar" that represents the NOT operation – particularly in drop-down boxes (either when looking at the drop-down or after selecting one of the choices). But some browser/zoom combinations have trouble even with overbars in normal text.

Do not use Chrome on a Mac. Use Safari if you use a Mac.

Logical Operations

The book uses the words **AND**, **OR**, **XOR**, and **NOT** as the operators for Boolean logic equations. However, we most commonly represent these using operators, not words. The below table shows the operator symbols we will use in this class.

Logic Function	Equivalent Expressions		
NOT	NOT(Y)	$\bar{Y}$	
AND	Y AND Z	$Y \cdot Z$	YZ
NAND	$\overline{Y \text{ AND } Z}$	$\overline{Y \cdot Z}$	$\overline{YZ}$
OR	Y OR Z	$Y + Z$	
NOR	$\overline{Y \text{ OR } Z}$	$\overline{Y + Z}$	
XOR	Y XOR Z	$Y \oplus Z$	
XNOR	$\overline{Y \text{ XOR } Z}$	$\overline{Y \oplus Z}$	

Note: the required video discussed all of these logic functions. The book presents the **NOT**, **AND**, **OR**, and **XOR** logic functions in Chapter 2, and the **NAND** and **NOR** logic functions in Chapter 3.

Question 1

In Boolean logic, what value do we use to represent **FALSE**?

In Boolean logic, what value do we use to represent **TRUE**?

Correct answers:

1 0 2 1

Attempt 1 of 1 Score: 0/0 pts

NOT Function

Question 2.1

To express the **NOT** operation, we draw a line the operand.

Correct answers:

1 over

Attempt 1 of 1 Score: 0/0 pts

Question 2.2

The result of a **NOT** operation will be the the operand's value.

Correct answers:

1 always 2 opposite of

Question 2.3

Complete the below truth table for the **NOT** function based on the correct description of the function above.

Do not look up the truth table to fill it in — you should be able to fill it in based on understanding the function!

A	$\bar{A}$
0	1
1	2

Correct answers:

1 1 2 0

AND Function

Question 3.1

To express the function $Y = A \text{ AND } B$, we can also write $Y = A$

1	
---	--

 B .

Correct answers:

1 •

We actually usually write the AND function as $Y = AB$, with no explicit operator between the two variables, the same way we usually don't show the multiplication operator in normal algebra (e.g., $f(x,y) = xy$).

Question 3.2

The result of an **AND** operation will be **TRUE** if

1	
---	--

 of the operands are **TRUE**.

Correct answers:

1 all

Attempt 1 of 1 Score: 0/0 pts

Question 3.3

The result of an **AND** operation will be **FALSE** if

1	
---	--

 of the operands are **FALSE**.

Correct answers:

1 any

Attempt 1 of 1 Score: 0/0 pts

Question 3.4

Complete the below truth table for the **AND** function based on the correct descriptions of the function above.

Do not look up the truth table to fill it in — you should be able to fill it in based on understanding the function!

A	B	A·B	
0	0	1	
0	1	2	
1	0	3	
1	1	4	

Correct answers:

1 0 2 0 3 0 4 1

Attempt 1 of 1 Score: 0/0 pts

OR Function

Question 4.1

To express the function $Y = A \text{ OR } B$, we can also write $Y = A$

1	
---	--

 B .

Correct answers:

1 +

Attempt 1 of 1 Score: 0/0 pts

In **Boolean** algebra, the "plus" sign means **OR**, not addition! So, don't say "Y equals A plus B" – say "**Y equals A OR B**". If you say "plus" you're more likely to confuse yourself or others...

Question 4.2

The result of an **OR** operation will be **TRUE** if

1	
---	--

 of the operands are **TRUE**.

Correct answers:

1 any

Attempt 1 of 1 Score: 0/0 pts

Question 4.3

The result of an **OR** operation will be **FALSE** if

1	
---	--

 of the operands are **FALSE**.

Correct answers:

1 all

Attempt 1 of 1 Score: 0/0 pts

Question 4.4

Complete the below truth table for the **OR** function based on the correct descriptions of the function above.

Do not look up the truth table to fill it in — you should be able to fill it in based on understanding the function!

A	B	A+B	
0	0	1	
0	1	2	
1	0	3	
1	1	4	

Correct answers:

1 0 2 1 3 1 4 1

Attempt 1 of 1 Score: 0/0 pts

Boolean Operations

Note: If you can't remember a particular operator symbol, refer to the table at the start of the exercise.

Question 5.1

$$Y = \bar{1}$$

Note: this could also be written as $Y = \text{NOT } 1$

What is the value of the Boolean variable **Y**?

Correct answers:

0

Attempt 1 of 1 Score: 0/0 pts

Question 5.2

$$Y = 1 \cdot 1$$

Note: this could also be written as $Y = 1 \text{ AND } 1$

What is the value of the Boolean variable **Y**?

Correct answers:

1

Attempt 1 of 1 Score: 0/0 pts

Question 5.3

$$Y = 0 \cdot 1$$

Note: this could also be written as $Y = 0 \text{ AND } 1$

What is the value of the Boolean variable **Y**?

Correct answers:

0

Attempt 1 of 1 Score: 0/0 pts

Question 5.4

$$Y = 1 + 1$$

*Note: this could also be written as $Y = 1$ **OR** 1*

What is the value of the Boolean variable **Y**?

Correct answers:

1

Attempt 1 of 1 Score: 0/0 pts

Question 5.5

$$Y = 0 + 1$$

*Note: this could also be written as $Y = 0$ **OR** 1*

What is the value of the Boolean variable **Y**?

Correct answers:

1

Transistors

Question 6.1

In a digital circuit, the transistors behave like

1	
---	--

 that allow or prevent the flow of electricity.

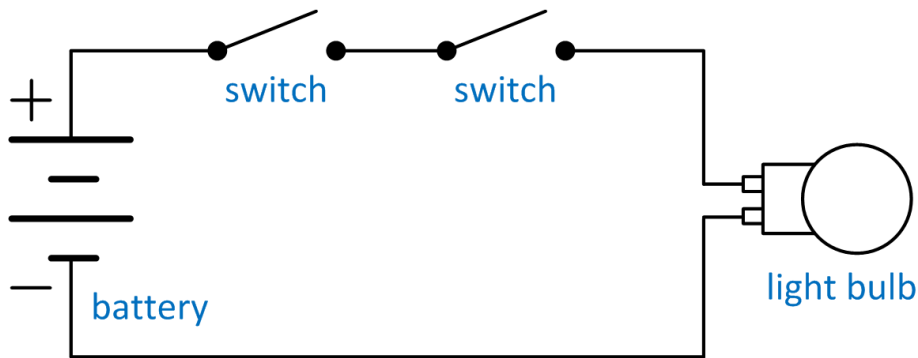
Correct answers:

1 switches

Attempt 1 of 1 Score: 0/0 pts

Question 6.2

Consider a very simple electric circuit consisting of a battery, a light bulb, and two switches, as shown below.



Assume that a switch being closed (on) is a logical 1 (TRUE), and the bulb lighting up is a logical 1 (TRUE).

If we think of the switches as the inputs and the light bulb as the output, what basic logical operation does this circuit implement?

- ☐ AND
- ☐ NOT
- ☐ OR

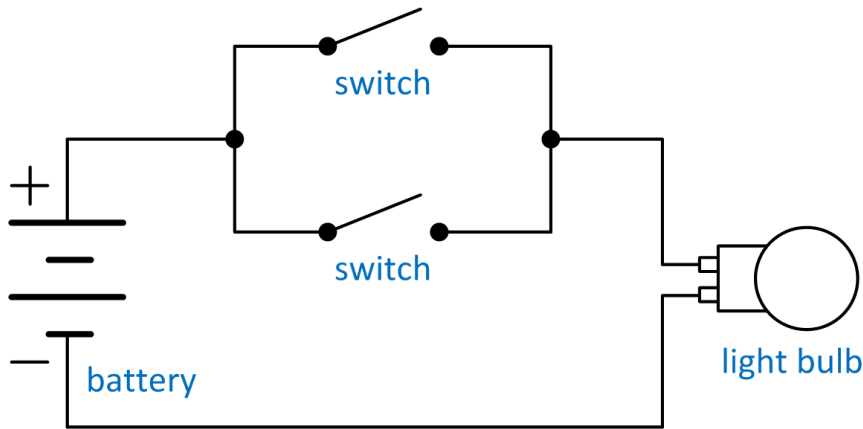
Attempt 1 of 1 Score: 0/0 pts

The light will only turn on if the left switch **AND** the right switch are closed, because that is the only way

The light will only turn on if the left switch **AND** the right switch are closed, because that is the only way current can flow to the light bulb.

Question 6.3

Consider a very simple electric circuit consisting of a battery, a light bulb, and two switches, as shown below.



Assume that a switch being closed (on) is a logical 1 (TRUE), and the bulb lighting up is a logical 1 (TRUE).

If we think of the switches as the inputs and the light bulb as the output, what basic logical operation does this circuit implement?

- ☐ NOT
- ☐ OR
- ☐ AND

Attempt 1 of 1 Score: 0/0 pts

The light will turn on if either the top switch **OR** the bottom switch is closed (or both!), because that will allow current to flow to the light bulb.

Question 7.1

In CMOS-based digital circuit design (i.e., using N-FETs and P-FETs), "power" (usually labeled V_{dd} or V_{CC}) represents logic level

1	
---	--

, whereas "ground" (GND) represents logic level

2	
---	--

.

Correct answers:

1 1 2 0

Attempt 1 of 1 Score: 0/0 pts

Question 7.2

Logic gates are simple electric circuits, built from transistors, that are designed to implement logic functions.

The logic circuit abstraction layer allows us to (for the most part) think of these circuits in terms of structures that manipulate

1	
---	--

 to perform computations instead of thinking in terms of circuits that manipulate voltage levels.

Correct answers:

1 Boolean values (0, 1)

Attempt 1 of 1 Score: 0/0 pts

Question 7.3

When we design circuits using logic gates, we're working at a

1	
---	--

 layer of abstraction than if we were to design our circuits transistor-by-transistor.

Correct answers:

1 higher

Attempt 1 of 1 Score: 0/0 pts

Question 7.4

How many transistors are needed to make a NOT gate (also called an inverter) using CMOS technology?

You may wish to refer to the book or videos.

Correct answers:

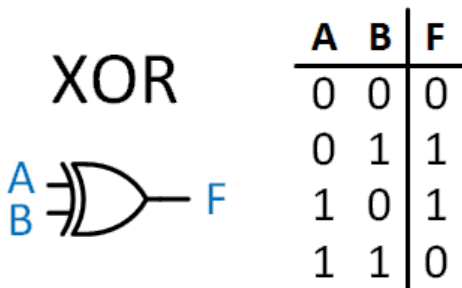
2

Attempt 1 of 2 Score: 0/0 pts

XOR and XNOR

Question 8.1

A very common gate type that was not mentioned in the textbook is the exclusive OR (**XOR**) gate. While this is not a fundamental gate, the exclusive OR operation is very commonly used in both hardware and software, so you need to know it. The gate symbol and truth table are shown below.



The **XOR** operation is also said to implement the function, because of the number of inputs that must be 1 in order to produce an output of 1.

Correct answers:

1 odd

Attempt 1 of 1 Score: 0/0 pts

Question 8.2

The 2-input **OR** gate is more precisely described as the **inclusive-OR**. It is given that name because in addition to the cases when only one input is true, it also **includes** the case when both inputs are true.

The 2-input **exclusive-OR**, or **XOR**, gate, is so named because it

 the case when both inputs are true, so the 2-input **XOR** operation is true when

.

Correct answers:

1 excludes 2 one and only one input is true

Attempt 1 of 1 Score: 0/0 pts

Question 8.3

$$Y = 1 \oplus 1$$

Note: this could also be written as $Y = 1 \text{ XOR } 1$

What is the value of the Boolean variable **Y**?

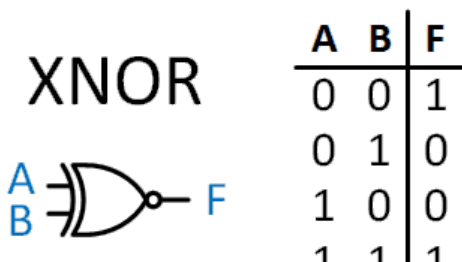
Correct answers:

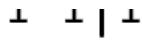
0

Attempt 1 of 1 Score: 0/0 pts

Question 8.4

The **XNOR** gate is closely related to the **XOR** gate. The **XNOR** gate symbol and truth table are shown below.





Compare the **XOR** and **XNOR** truth tables. Based on that, you can surmise that the 'N' in **XNOR** stands for the

1 function.

Correct answers:

1 NOT

Attempt 1 of 1 Score: 0/0 pts

Question 9

Drag-and-drop the appropriate logic gate symbol to match each gate name below.

1

2

3

4

5

OR
NOT
AND
XOR
NAND

Correct answers:

1 2 3 4 5

Attempt 1 of 2 Score: 0/0 pts

NAND and NOR

Question 10.1

The **NAND** function has the same name as the **AND** function except for the letter

1	
---	--

 at the start of the name.

In the truth table of the **NAND** function, the result in each row is

2	
---	--

 the **AND** function.

Correct answers:

1 N 2 the opposite of

Attempt 1 of 2 Score: 0/0 pts

Question 10.2

The **NOR** function has the same name as the **OR** function except for the letter

1	
---	--

 at the start of the name.

In the truth table of the **NOR** function, the result is

2	
---	--

 the **OR** function.

Correct answers:

1 N 2 the opposite of

Attempt 1 of 2 Score: 0/0 pts

Remember, the 'N' at or near the start of the names **NAND**, **NOR**, and **XNOR** means **NOT**.

Question 10.3

All of the gates that have the letter 'N' at or near the start of their name (**NAND**, **NOR**, **XNOR**, and **NOT**) also

1	
---	--

Correct answers:

- 1 have a "bubble" (a small circle) on the output of their gate symbol

Attempt 1 of 1 Score: 0/0 pts

In the questions on the next several pages, determine the output of a logic gate based on the type of gate and its input(s).

Gate input(s) may be connected to constant values of 0 or 1, or may be connected to a wire carrying a signal named **K**. The **K** signal will be equal to 0 or it will be equal to 1 at any given time, but you don't know which at this moment, so you have to treat the **K** signal value as a variable of the same name.

HINT: If you are not sure how to determine the output, try making a truth table based on the logic function implemented by the gate... If you don't understand what to do, get help from a neighbor or an instructor!

Question 11



The output of the above gate will be equal to...

☐ 0

☐ 1

Attempt 1 of 1 Score: 0/0 pts

Question 12



The output of the above gate will be equal to...

- ☐ 0
- ☐ 1
- ☐ K
- ☐ $\bar{K}$

Attempt 1 of 2 Score: 0/0 pts

Question 13



The output of the above gate will be equal to...

- ☐ 0
- ☐ 1
- ☐ K
- ☐ $\bar{K}$

Attempt 1 of 2 Score: 0/0 pts

Question 14



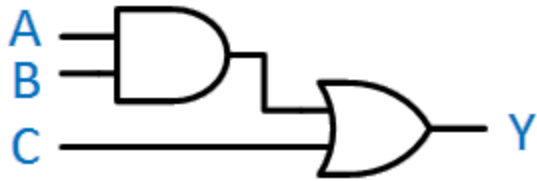
The output of the above gate will be equal to...

- ☐ 0
- ☐ 1
- ☐ K



Attempt 1 of 2 Score: 0/0 pts

Question 15



Complete the below truth table for the logic function described by the above circuit.

A	B	C	Y	
0	0	0	1	
0	0	1	2	
0	1	0	3	
0	1	1	4	
1	0	0	5	
1	0	1	6	
1	1	0	7	
1	1	1	8	

Correct answers:

1 0 2 1 3 0 4 1 5 0 6 1 7 1 8 1

Attempt 1 of 1 Score: 0/0 pts

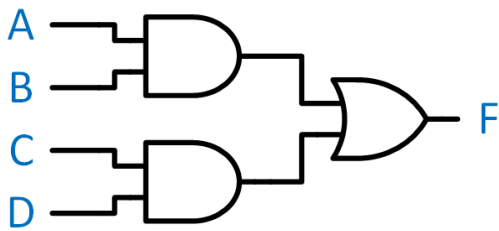
In normal algebra, there is an established operator precedence order so we know that $7 \times 2 + 3 = 17$ and $7 \times (2 + 3) = 35$.

In Boolean algebra, the precedence order is **parentheses**, **AND**, and then **OR**.

A complement bar (**NOT**) acts as if it has implicit parentheses, so $\overline{A+BC}$ is evaluated as if it were $\overline{(A+(BC))}$.

Question 16

Consider the logic circuit shown below.



Which Boolean function directly represents the Boolean logic implemented by the circuit?

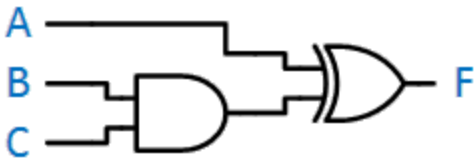
1	
---	--

Correct answers:

1 $F = AB + CD$

Attempt 1 of 2 Score: 0/0 pts

Question 17



Complete the below logic function that describes the above gate diagram.

F =	1		2		3	
-----	---	--	---	--	---	--

Correct answers:

1 A 2 $\oplus$ 3 $(B \cdot C)$

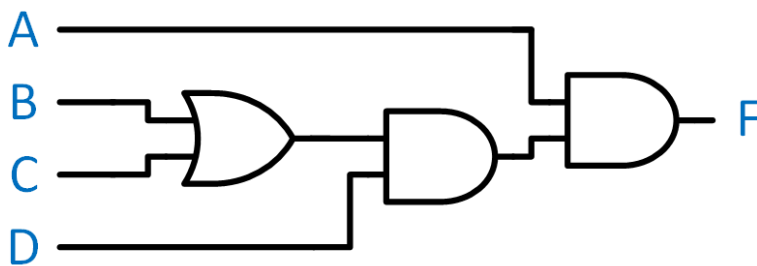
Attempt 1 of 2 Score: 0/0 pts

In an equation, parentheses are used to indicate precedence of operations.

The equation should perform same operations on the same signals and in the same order as the circuit.

Question 18

Consider the logic circuit shown below.



Which Boolean function directly represents the Boolean logic implemented by the circuit?

1	
---	--

Correct answers:

1 $F = A((B + C)D)$

Attempt 1 of 2 Score: 0/0 pts

Waveforms

Question 19

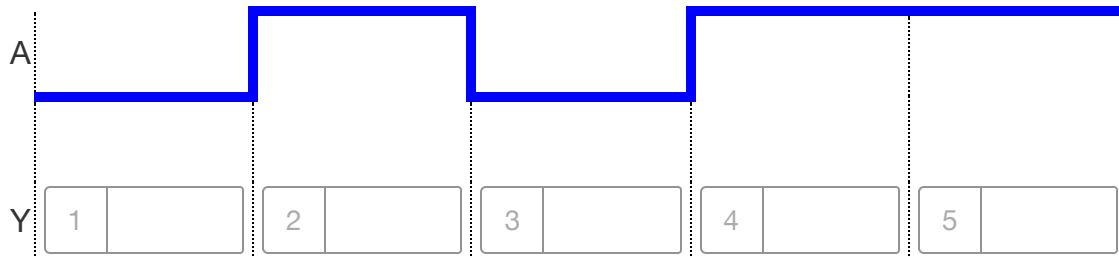
The input of a inverter (NOT gate) is connected to signal **A**, and its output is connected to signal **Y**, as shown below.



A waveform diagram is simply a graph where time is represented on the horizontal axis, and the values of multiple logic signals are plotted on the vertical axis. A "low" line segment on the waveform indicates that the signal is 0 during that time, and a "high" line segment indicates that the signal is 1 during that time.

The waveform below shows that signal **A** (the input to the inverter) is logic 0 in the first time interval, logic 1 in the next time interval, logic 0 in the third interval, and then it is 1 for the last two time intervals.

Complete the waveform by selecting whether the inverter's output **Y** is 0 or 1 in each time interval.



Correct answers:

1 1 2 0 3 1 4 0 5 0

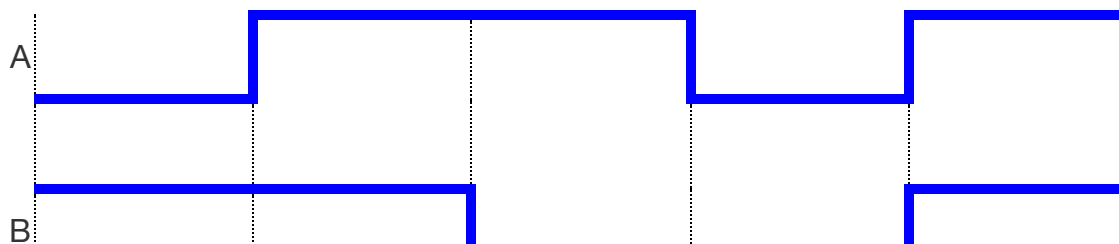
Attempt 1 of 1 Score: 0/0 pts

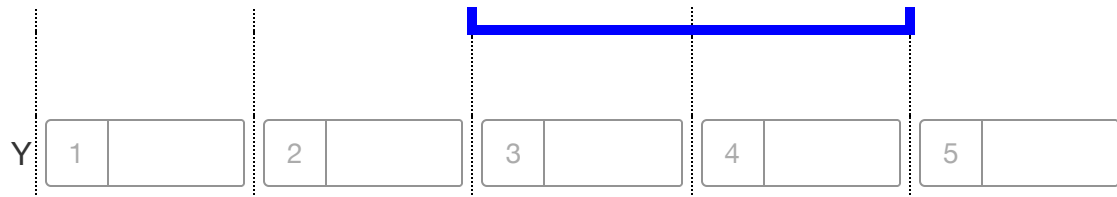
Question 20

The inputs of a logic gate are connected to signals **A** and **B**, and its output is connected to signal **Y**, as shown below.



Complete the waveform by selecting whether the output **Y** is a 0 or 1 in each time interval.





Correct answers:

1 0 2 1 3 0 4 0 5 1

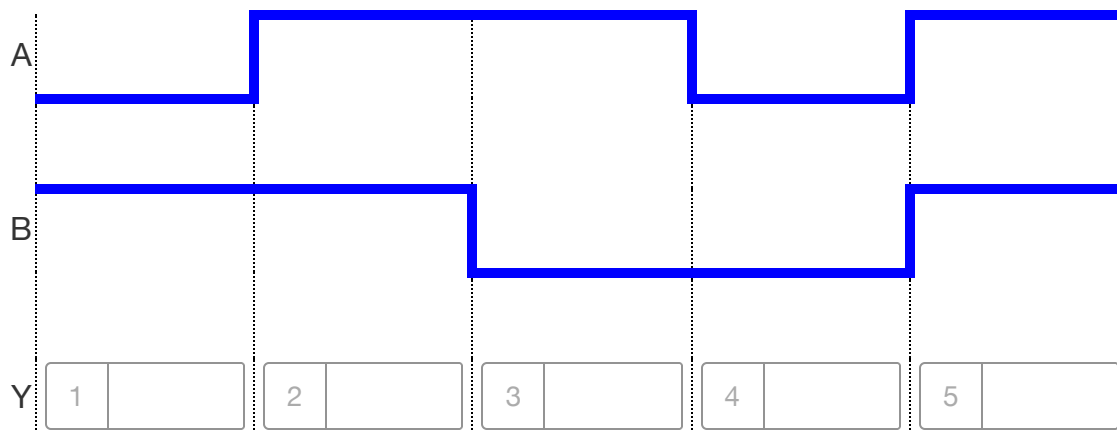
Attempt 1 of 1 Score: 0/0 pts

Question 21

The inputs of a logic gate are connected to signals **A** and **B**, and its output is connected to signal **Y**, as shown below.



Complete the waveform by selecting whether the output **Y** is a 0 or 1 in each time interval.



Correct answers:

1 1 2 0 3 1 4 0 5 0

Attempt 1 of 1 Score: 0/0 pts

Bitwise Logic Operations

Question 22.1

What does it mean if you perform a "bitwise" logical operation? Choose the **best** answer.

- ☐ The result is a single bit.
- ☐ Only some of the bits of the operands are processed.
- ☐ Some bits are "wiser" than others due to more life experience.
- ☐ For each bit of the result, we apply the operation to the corresponding bit(s) of the operand(s).

Attempt 1 of 1 Score: 0/0 pts

Question 22.2

You are given two 4-bit binary numbers:

1101

0100

What is the result of a bitwise **XOR** (exclusive OR) of these two numbers?

Enter your answer as a **binary** number.

Correct answers:

1001

